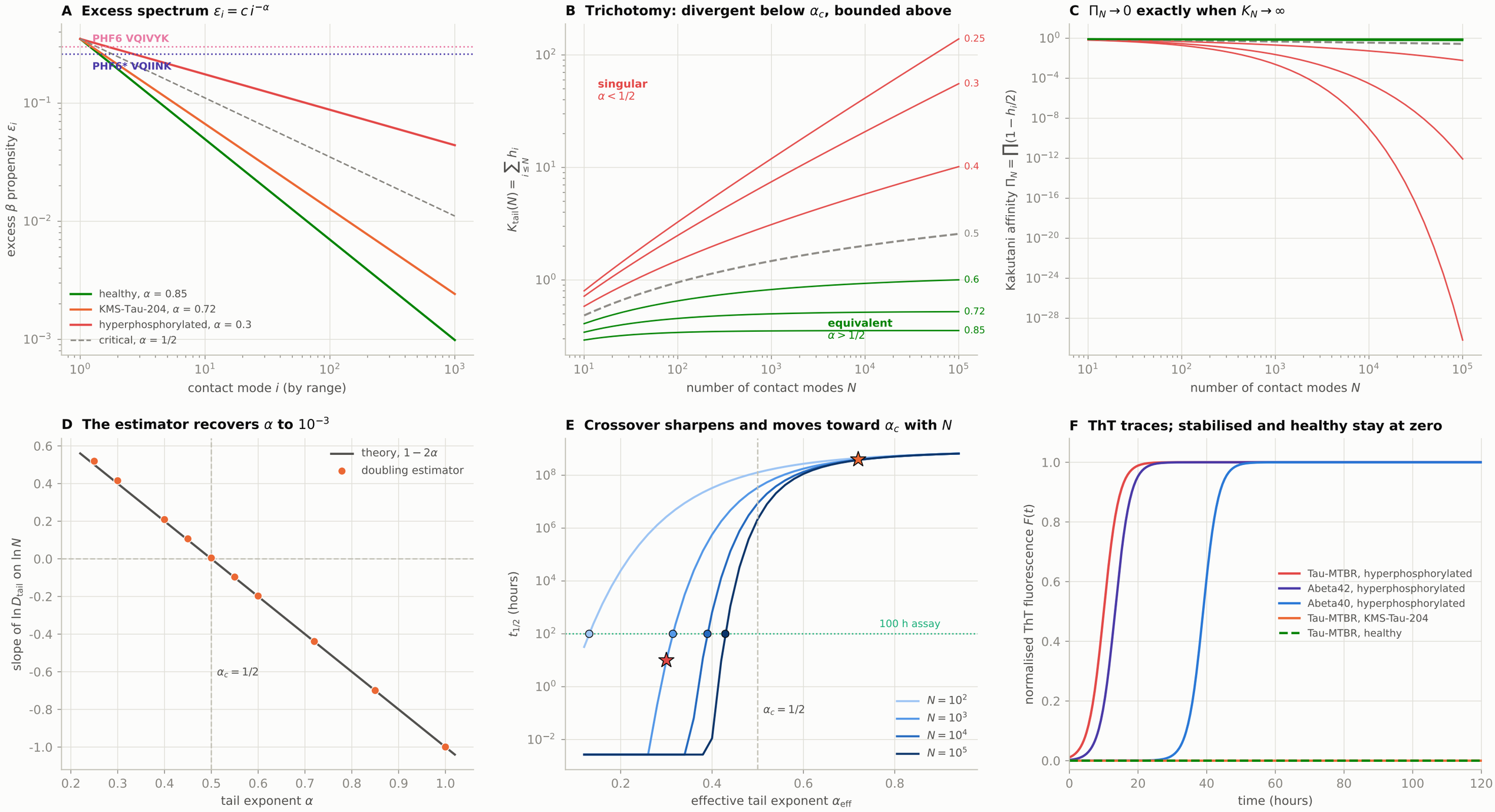

A Excess spectrum $\varepsilon_i = c\, i^{-\alpha}$
PHF6 VQIVYK
PHF6* VQIINK
excess β propensity ε_i
contact mode i (by range)
healthy, α = 0.85
KMS-Tau-204, α = 0.72
hyperphosphorylated, α = 0.3
critical, α = 1/2
B Trichotomy: divergent below α_c, bounded above
singular
$\alpha < 1/2$
equivalent
$\alpha > 1/2$
$K_{\mathrm{tail}}(N) = \sum_{i \le N} h_i$
number of contact modes N
0.25
0.3
0.4
0.5
0.6
0.72
0.85
C $\Pi_N \to 0$ exactly when $K_N \to \infty$
Kakutani affinity $\Pi_N = \prod (1 - h_i/2)$
number of contact modes N
D The estimator recovers α to 10^{-3}
theory, $1 - 2\alpha$
doubling estimator
slope of $\ln D_{\mathrm{tail}}$ on $\ln N$
$\alpha_c = 1/2$
tail exponent α
E Crossover sharpens and moves toward α_c with N
$t_{1/2}$ (hours)
100 h assay
$\alpha_c = 1/2$
$N = 10^2$
$N = 10^3$
$N = 10^4$
$N = 10^5$
effective tail exponent α_{eff}
F ThT traces; stabilised and healthy stay at zero
normalised ThT fluorescence $F(t)$
Tau-MTBR, hyperphosphorylated
Abeta42, hyperphosphorylated
Abeta40, hyperphosphorylated
Tau-MTBR, KMS-Tau-204
Tau-MTBR, healthy
time (hours)